

Learning by Correcting AI Errors: Effort is Essential

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Abstract. Concerns about declining student comprehension and critical thinking skills have led to new pedagogies based on fact-checking generative AI. However, it is unclear whether these pedagogies foster learning for all students. This study investigated a simplified scenario for humans fact-checking AI where participants corrected a virtual student. In two experiments, participants took a pre-test and then participated in three conditions: read-only (Read), read with an erroneous virtual student (Err), and read with a correct virtual student (Cor), followed by a post-test. Experiment 1 found that participants from an undergraduate subject pool ($N = 92$) learned more in Cor than Err, $d = .38$, and more in Cor than Read, $d = .37$, but that no learning occurred in Err and Read conditions. Experiment 2 found that participants from Amazon Mechanical Turk ($N = 85$) learned more from the Err than Read, $d = .72$, but that Cor was not significantly different from Err or Read. Follow-up analyses suggest that participants in the two experiments exhibited drastically different correcting behaviors: only 52% of undergraduates corrected the virtual student on selected tasks, whereas 98% of crowdworkers corrected the virtual student on the same tasks. Mediation analysis indicates that for crowd workers, learning was entirely mediated by their correcting behavior. Altogether, these results suggest that learning by correcting errors can be effective but only if students put in the effort.

Keywords: reading comprehension · critical thinking · learning by teaching · teachable agent

1 Introduction

The importance of critical thinking cannot be overstated. Critical thinking has been called an essential 21st-century skill in multiple learning frameworks [2, 19, 22]. A large survey of employers found that critical thinking was among the top five most important skills for job success across education levels [4]. Economic analyses have found that critical thinking skills reliably predict economic growth and are more predictive of economic growth than years spent in school [10]. However, recent reviews of large language models in education have noted the potential for large language models (LLMs), like ChatGPT, to misinform students and degrade critical thinking skills, and have lead to recommendations to integrate critical thinking and fact-checking activities into the usage of LLMs [8, 13, 16, 18]. These recommendations align with a growing movement of educators who argue that students should be allowed to use LLMs in their school work because using AI is an important job skill [1].

Studies have begun to integrate critical thinking and fact-checking activities into classroom use of LLMs. A common task across most of these studies is to have students use an LLM to write an essay and then fact-check sources, both in terms of whether they exist and whether they support the claim made in the text [12]. However, the educational outcomes from integrating critical thinking and fact-checking activities have been mixed. One case study asked undergraduate students in a chemistry class to engage in LLM essay fact-checking, with two additional steps: correcting the LLM essay by finding new references if incorrect ones were used and by revising claims that could not be supported by references and reasoning [9]. Students were not trained to perform these tasks, and the study found that some students were able to find and correct misinformation and irrelevant information, but other students were not. A high school case study used similar tasks of generating essays and answers that students fact-checked, critiqued, and improved upon, but LLM use was highly structured and integrated into classroom activities over the course of a semester [5]. The instructor in this study provided demonstrations of how to use the LLM for these tasks and provided feedback to students on how to use the LLM more effectively. This study concluded that students were able to successfully fact-check the LLM but that students found fact-checking very difficult.

Perhaps the best results in using LLMs to foster critical thinking have been achieved in a one-year randomized controlled trial on LLM-supported debate practice [23]. Two groups of college students engaged in debate sessions once a week for 90 minutes. One group used an LLM both to generate multiple viewpoints on the debate topic, which the students fact-checked, and to generate counterarguments to their own as a live debate partner. Compared to controls who did not use LLMs, students in the LLM group showed significant large effects on pre-post gain for critical thinking in international relations, including understanding of complex concepts and argumentation skills.

Critical thinking involves multiple skills, so it is perhaps not surprising that longer-duration studies involving both training and practice had improved outcomes. Critical thinking has four component skills that are generally agreed upon: analyzing arguments, claims, or evidence; making inferences; judging or evaluating; and making decisions or solving problems [14]. Reasoning, problem-solving, and decision-making require comprehension as a first step [17], and specifically reading comprehension when it comes to reading LLM-generated essays. By first comprehending text, readers construct a mental model that includes the claims and evidence in a text, at which point critical thinking processes like analyzing, evaluating, and decision-making can occur. The essay fact-checking tasks described above introduce additional complexity by requiring students to consult other sources to verify claims and correct misinformation by consulting multiple texts. Reading comprehension that spans multiple texts is known as multiple document comprehension [3], i.e. students must use a search strategy to find new sources, analyze and evaluate multiple sources, and engage in critical thinking to decide what information should be included, revised, or deleted.

When students fail on LLM essay fact-checking tasks, it is unclear where the breakdown occurred, i.e. whether it occurred in comprehension, search for alternative documents, analysis of claims, etc. We conducted a series of experiments in 2015 that simplified classroom fact-checking studies in multiple ways. First, reading was reduced

to a single source, a short text that was always on screen. Participants therefore needed to construct only a small mental model and could easily glance at the text to refresh their memory. Second, participants received precisely controlled correct and incorrect information on concept maps, so the ideal correcting behavior was experimentally determined. Finally, the AI in the study, presenting as a virtual student, enacted reading comprehension skills, based on reciprocal teaching [21], and explained its reasoning. The agent thus scaffolded participant reading comprehension to support weaker comprehenders. The main goal of our experiments was to explore whether humans could generate training data for AI by correcting a virtual student, but the results were complex and never published. We believe these experiments, by exploring a simplified fact-checking scenario, can provide insight into pedagogies for fact-checking LLMs that require a more complex integration of skills.

In this paper, we present the learning analysis of those experiments, specifically how learning is affected when the virtual student never makes mistakes vs. when the virtual student makes mistakes which the humans can then correct. In these experiments, the virtual students are a proxy for LLMs; however, their actions are pre-scripted to maintain experimental control. Our research questions are 1) Do humans learn more with a virtual student than by reading the text; 2) Do humans learn more with the erroneous virtual student who makes mistakes or the correct virtual student who never makes mistakes? We hypothesized that humans would learn more from a virtual student than reading the text alone, because the enacted reading comprehension strategies based on reciprocal teaching should help humans construct a more accurate mental model of the text. We additionally hypothesized that humans who corrected virtual student mistakes would learn the most, because fact-checking would lead to deeper processing and a more durable memory of the text [7, 11].

2 Experiment 1

2.1 Method

Design. The study used a repeated measures design in which participants took a pre-test and completed three conditions, read-only (Read), read with an erroneous virtual student (Err), and read with a correct virtual student (Cor), followed by a post-test. The Read condition always came first, but the order of Err and Cor was counterbalanced so some participants experienced Read-Err-Cor and other participants experienced Read-Cor-Err. Each of the three conditions covered one of three topics, and topics were similarly counterbalanced across all possible orders. Counterbalancing of conditions and topics was performed to minimize potential order effects; pre- and post-test were also counterbalanced to avoid order effects and minor differences in difficulty between tests. The primary outcome measures for the study were 1) simple learning gains (post-test minus pre-test score) and 2) correction behavior. Because a wide range of correction behaviors was observed, for the purpose of this paper we collapse correction behavior down to a binary indicator: did the participant make at least one change on a majority of concept maps or not? Because the design is repeated measures, data were analyzed using mixed models.

Participants. Undergraduate students ($N = 102$) at an urban university in the U.S. volunteered to participate in the study in 2015. The average age was 21.04 years ($SD = 4.67$, range: 18-47 years). A total of 76 female and 26 male participants were recruited. Participants were members of a Psychology Department subject pool and so received either extra credit for participation or course assignment credit, where participation was an alternative to completing a course assignment.

Materials Participants used a web browser to complete the experiment remotely. The primary materials used in the study were a web-delivered learning environment and pre- and post-tests. The learning environment was constructed in Microsoft Silverlight and was designed as a textbook in the Read condition and as an intelligent textbook in the other conditions, as shown in Figure 1. The textbook component covered three psychology topics, namely, attitudes, affiliation, and antisocial behavior, with text adapted from an introduction to psychology textbook [6]. Each topic was displayed across seven pages of text containing approximately 120-180 words each, such that the entire text could be read without scrolling on a screen larger than 640x480 pixels. The intelligent textbook conditions launched a virtual student after the text was read. The virtual student used the Media Semantics platform and presented as a talking head with synthesized speech. The virtual student enacted the following reading comprehension skills for each page of text: gist (what the text is about), question generation, concept mapping, and prediction (what the next page will be about). These reading comprehension skills were modeled on reciprocal teaching [21], a multi-strategy approach to improving reading comprehension that has been shown in various meta-analyses to be effective [24, 25]. We adapted reciprocal teaching tasks to be better aligned with the goal of creating AI training data [20] such that the standard reciprocal teaching tasks of questioning, clarifying, and summarizing were realigned with the concept mapping task, and the other tasks of gist/summarization and question generation were thus more superficial than their reciprocal teaching counterparts. The virtual student asked for correction after enacting each reading comprehension skill. Correction took place by directly editing what the virtual student produced, either by typing in a text box or interactively editing the concept map, i.e. editing nodes/links or adding new nodes/links.

In the Cor condition, the virtual student never made mistakes: all Cor answers were consistent with the text and were checked for accuracy by the authors of this paper. The Err condition used a version of the Cor condition concept map where the map was altered according to the following rules based on human errors in reading comprehension [17]: 1) if a node is not in the prior sentence, create a new node (a dis-integration rule: working memory); 2) if a node refers to an existing entity, but uses a different label, then create a new node (a dis-integration rule: working memory/vocabulary/prior knowledge); 3) never resolve coreference (a deletion rule: working memory); 4) never have a correct elaborative inference (a deletion rule: prior knowledge); 5) never use a label that requires inference (a substitution rule: prior knowledge); 6) contradict one statement in the text (a negation rule: test of situation model and monitoring); 7) insert one topically related but irrelevant item external to the text (an addition rule: test of situation model and monitoring); 8) insert one irrelevant item internal to the text (an addition rule: test of situation model and monitoring). In summary, rules 1-5 resulted in a poorer quality

Social Cognition: Attitudes and Attitude Change-Copping a 'Tude

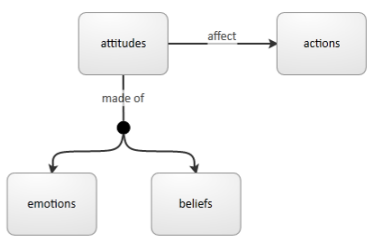
Journey Question 14.3 How are attitudes acquired and changed?

What is your attitude toward affirmative action, euthanasia, environmental groups, the situation in the Middle East, the death penalty, legalized abortion, junk food, psychology? Your answers, which are often influenced by social situations, can have far-reaching effects on your behavior. Attitudes are intimately woven into our actions and views of the world. Our tastes, friendships, votes, preferences, goals, and behavior in many other situations are all touched by attitudes (Baumeister and Bushman, 2011). Let's see how attitudes are formed and changed.

What specifically is an attitude? An attitude is a mixture of belief and emotion that predisposes a person to respond to other people, objects, or groups in a positive or negative way. Attitudes summarize your evaluation of objects (Bohner and Dickel, 2010). As a result, they predict or direct future actions.



Alright. So the important things to remember are Attitudes are made of beliefs Attitudes are made of emotions Attitudes affect actions Does that sound right?



Submit

Fig. 1. Learning environment during the concept mapping task (Cor condition)

concept map, rule 6 created an incorrect fact in the concept map, and rules 7-8 added irrelevant information to the concept map. Virtual student performance on the other reading comprehension tasks was based on the concept maps, so concept maps were the primary difference between the Cor and Err conditions.

Two instructional videos were included showing how to use the learning environment, one for the Read condition and one for the virtual student conditions. The 30-second Read condition video explained how to use the interface for the reading task and included instructions for remembering as much of the text as possible because there would be a test later. The 5-minute virtual student condition video explained how to use the interface for the virtual student tasks, including how to correct the virtual student for each task. Approximately 3 minutes of the training video focused on how to use the concept map, including how to read the concept map and how to edit it by creating nodes, editing the text on nodes, creating links, editing the text on links, deleting nodes and links, and moving nodes and links around.

The pre- and post-tests were created by the authors to match the information in each text. Ten questions from each topic were created and then parallel questions of each were created to make another form of the same test. The questions were piloted in a read-only condition where one form of the test was given, participants read all three topic texts, and then the other form of the test was given. Participants ($N = 59$) were randomly assigned to an order of test forms, i.e. either form A followed by form B or form B followed by form A. Analysis of the forms suggested that some questions were excessively easy or difficult, so these questions were removed from the final tests. Form A of the final test consisted of 22 multiple-choice questions, with 6, 7, and 9 questions

from attitude, affiliation, and antisocial behavior topics respectively. Form B of the final test consisted of 23 multiple-choice questions, with 7, 8, and 8 questions from the same respective topics. All tests were delivered through the Qualtrics web platform.

Procedure. Participants entered the study by clicking on a Qualtrics link. After completing informed consent, participants were randomized to condition order, either Read-Err-Cor or Read-Cor-Err, randomized to one of six topic orders, which represented all combinations of three topic orders, and randomized to one of two test orders, either form A followed by form B or form B followed by form A. After randomization, participants answered some exploratory general reading comprehension questions and took the pretest according to their randomized form. Participants then clicked on a link that opened the learning environment in another tab. The learning environment presented conditions and topics according to their randomly assigned order. Before the Read condition, participants watched an instructional video on how to use the interface for the reading task. During the Read condition, participants read seven pages of text, advancing to each text by clicking on a “Done Reading” button. Before the first virtual student condition, participants watched an instructional video on how to use the interface for the virtual student tasks. During each virtual student condition, when the participant pressed the “Done Reading” button, the virtual student would appear and enact the gist, question generation, concept map, and prediction skills in that order. Participants were able to correct the virtual student if they chose, and they pressed a “Submit” button to proceed to the next reading comprehension task. At the end of the tasks, the learning environment proceeded to the next page of text, and the process repeated itself until participants had read all texts and advanced through all tasks. Following the learning environment, participants took the post-test form according to their randomization order, some exploratory attitude and locus of control questions, followed by demographic questions.

2.2 Results & Discussion

Inspection of experimental session times indicated some implausibly long times, with the maximum session time being approximately 156 hours. Long session times could unfavorably advantage the most recently completed conditions due to forgetting between the first condition and the post-test. We decided to use the interquartile range * 1.5 (i.e. past boxplot whiskers) to identify session time outliers rather than setting an arbitrary threshold for session length. Removed outliers comprised 9.8% of the data, and the resulting session times had a maximum value of 3.62 hours. Of the participants enrolled ($N = 102$), ten were removed for having excessively long experimental session times, and the final sample ($N = 92$) was used for analysis. We present only results central to our research questions rather than results from exploratory measures due to page limitations.

Table 1 shows learning gains and concept map change behaviors across conditions. Learning gains for conditions other than Cor trended towards zero. Additionally, just over half of the participants made any change whatsoever to the majority of concept maps in the Err condition, all of which were guaranteed to have at least one incorrect

Table 1. Experiment 1 Descriptive Statistics For Learning and Corrections

Measure	Condition					
	Read		Err		Cor	
	M	SD	M	SD	M	SD
Learning Gains	.00	.23	-.01	.26	.08	.28
Majority Maps Changed			.54	.50	.38	.49

fact and multiple irrelevant facts. In contrast, over a third of participants made changes to the majority of maps in the Cor condition, all of which were correct. To determine if differences in learning gains were reliable, we created a linear mixed model and conducted statistical tests at $\alpha = .05$ to answer our research questions. The model with fixed effects was $LearningGain \sim Condition$, where Condition was Read, Err, or Cor. The model random effects necessarily include participant as there are multiple test responses per participant; however, we were unable to include topic as a random effect due to the low number of topics (3). Additionally, we were unable to use Condition as a random slope because then the number of observations would equal the number of random effects.

To answer our two research questions of whether participants learn more with a virtual student than reading the text and which virtual student participants learn more from, we conducted an ANOVA analysis of the mixed model [26]. ANOVA analysis revealed a significant main effect of Condition, $F(2, 182) = 4.26, p = .016$. We conducted contrasts within Condition using Tukey’s p-value adjustment [15]. Contrasts of learning gains between conditions revealed that learning gains in the Cor condition ($M = .08, SE = .03$) were higher than the Err condition ($M = -.01, SE = .03$), $t(182) = 2.56, p = .030, d = .38$, and also higher than the Read condition ($M = -.00, SE = .03$), $t(182) = 2.49, p = .036, d = .37$, but there was no difference between the Read and Err conditions, $t(182) = -.07, p = .997$. Noting the learning gain means for Read and Err were close to zero, we conducted one-sample t-tests to determine if the mean learning gains in these conditions were reliably different from zero. One sample t-tests revealed that learning gains for Read were not significantly different from zero, $t(91) = -.18, p = .860$, and that learning gains for Err were not significantly different from zero, $t(91) = -.24, p = .811$.

We were surprised that participants did not learn from reading the text or from correcting the Err virtual student. We questioned whether participants were fully compliant with the Err condition because only 50 out of 92 participants changed more than half of the concept maps in the Err condition. Thus it is possible that a learning effect for the Err condition could be masked by participants who did not comply with the experimental instructions. To test this compliance hypothesis, we conducted a mediation analysis using mixed models [27]. The mediating variable was MajorityMapChange, i.e. whether the participant had made any change to a majority of concept maps, and the corresponding logistic model with fixed effects was $MajorityMapChange \sim Condition$, where Condition was Err or Cor (i.e., there are no maps the Read condition). The model predicting learning with fixed effects was $LearningGain \sim MajorityMapChange + Condition$, where Condition was again Err or Cor. For both models, the random effect was par-

ticipant. Analysis of the model revealed that while Condition significantly predicted whether participants changed more than half of their concept maps in the Err condition, $z = 2.20$, $p = .028$, $OR = 1.29$, the mediated effect of MajorityMapChange on learning gain was not significant, $p = .298$.

Altogether, our results indicate that these undergraduate participants learned only from the Cor virtual student, i.e. the virtual student that was always correct. The lack of learning in the other conditions suggests that students were unwilling to read the text or correct the Err virtual student. Thus our hypothesis was partially supported: participants learned more from a virtual student than from reading the text alone, but participants did not learn the most from correcting an erroneous virtual student. We cannot fully reject the possibility that these undergraduate participants were unable to read and comprehend the text itself, but this alternative explanation seems unlikely given their presumed reading ability as college students and the non-technical nature of the text, which had a Flesch-Kincaid grade level of 10.60. Instead, we hypothesize that the undergraduate participants were not putting sufficient effort into the task.

3 Experiment 2

We decided to replicate the first experiment with a different population to test if the effects of Experiment 1 would generalize. All aspects of the study design were preserved except the participants were sampled from a different population.

3.1 Method

Participants. Participants ($N = 88$) from Amazon Mechanical Turk (AMT) were recruited to participate in the study in 2015. Participants were required to have an approval rating on AMT of at least 95% and to have completed at least 500 tasks on AMT. The average age was 33.23 years ($SD = 10.40$, range: 19-63 years). A total of 48 female and 40 male participants were recruited. Participants were compensated \$10.

3.2 Results & Discussion

Inspection of experimental session times again revealed implausibly long times, with the maximum session time being approximately 8.45 hours. To avoid learning gains bias due to forgetting earlier conditions, we used the interquartile range * 1.5 (i.e. past boxplot whiskers) to identify session time outliers rather than setting an arbitrary threshold for session length. Removed outliers comprised 3.4% of the data, and the resulting session times had a maximum value of 3.65 hours. Of participants enrolled ($N = 88$), 3 were removed for having excessively long experimental session times, and the final sample ($N = 85$) was used for analysis. We present only results central to our research questions rather than results from exploratory measures due to page limitations.

Table 2 shows learning gains and concept map change behaviors across conditions, with two notable trends. First, means for both virtual student conditions trend higher than the Read condition. Second, almost all participants made a change to the majority of concept maps in the Err condition, all of which were guaranteed to have at least one

Table 2. Experiment 2 Descriptive Statistics For Learning and Corrections

Measure	Condition					
	Read		Err		Cor	
	M	SD	M	SD	M	SD
Learning Gains	.12	.26	.22	.27	.19	.26
Majority Maps Changed			.98	.15	.52	.50

incorrect fact and multiple irrelevant facts. Additionally, over half of the participants made changes to the majority of maps in the Cor condition, all of which were correct. To determine if differences in learning gains were reliable, we created a linear mixed model and conducted statistical tests at $\alpha = .05$ to answer our research questions, with model formula and random effects matching those of Experiment 1.

To answer our two research questions of whether participants learn more with a virtual student than reading the text and which virtual student participants learn more from, we conducted an ANOVA analysis of the mixed model [26]. ANOVA analysis revealed a significant main effect of Condition, $F(2, 168) = 4.10, p = .018$. We conducted contrasts within Condition using Tukey’s p-value adjustment [15]. Contrasts of learning gains between conditions revealed that learning gains in the Err condition ($M = .22, SE = .03$) were higher than the Read condition ($M = .12, SE = .03$), $t(168) = 2.76, p = .018, d = .72$. However, the Cor condition ($M = .19, SE = .03$) was not significantly different from the Read condition, $t(168) = 2.05, p = .104$, or from the Err condition, $t(168) = .71, p = .757$. A one-sample t-test revealed that learning gains for Read were significantly different from zero, $t(84) = 4.25, p < .001$.

We conducted a mediation analysis using mixed models [27] to determine if correcting behavior in the Err condition could explain learning gains. The mediation model had the same structure as in Experiment 1: the mediating variable was MajorityMapChange, i.e. whether the participant had made any change to a majority of concept maps, and the logistic model with fixed effects was $MajorityMapChange \sim Condition$, where Condition was Err or Cor (i.e., there are no maps the Read condition). Likewise, the model predicting learning with fixed effects was $LearningGain \sim MajorityMapChange + Condition$, where Condition was again Err or Cor, and for both models, the only random effect was participant. Mediation analysis indicated that the effect of assignment to the Err condition on learning gains was entirely mediated by MajorityMapChange, $p = .014$, such that there was no direct effect of assignment to the Err condition on learning gains, $p = .442$. Inspection of the component models revealed that Condition significantly predicted whether participants changed more than half of their concept maps in the Err condition, $z = 4.47, p < .001, OR = 49.20$, and that MajorityMapChange further predicted learning gains, $t(165.35) = 2.31, p = .024, d = .53$.

Altogether, our results indicate that these crowd worker participants learned in all conditions, but the only condition with more learning than Read was the Err condition. Crowdworkers were highly compliant with the Err condition, with 83 out of 85 changing more than half of the concept maps in the Err condition. Learning in the Err condition was entirely mediated by whether participants changed the majority of concept maps, i.e. by their correcting behavior — no learning occurred unless they corrected the erroneous

virtual student. Thus our hypothesis was partially supported: participants learned more from a virtual student than from reading the text alone, but this applied only to the Err condition. It is possible that with a greater number of participants, and thus a greater power, the Cor condition might have significantly higher learning gains than the Read condition. However, the trend of learning gains in this study makes it seem unlikely that the Err condition will have higher learning gains than the Corr condition, even with more participants. We conclude that the effort that crowd worker participants put into correcting the erroneous virtual student paid off with higher learning.

4 General Discussion

The two experiments in this study examined a simplified scenario for humans fact-checking AI. Current pedagogies for fact-checking LLMs introduce additional complexity by requiring students to search for new sources, analyze and evaluate them, and engage in critical thinking to decide what LLM-generated information should be included, revised, or deleted. Instead of requiring students to search for multiple documents and read them to check claims, participants in our experiments read a single piece of text, which stayed on screen throughout the fact-checking process. The virtual students enacted reading comprehension skills, based on reciprocal teaching [21], and explained their reasoning, further simplifying the fact-checking task. Because all virtual student behaviors were tightly controlled, we could observe a clear contrast between learning with an erroneous virtual student who always produced some incorrect and irrelevant facts and learning with a correct virtual student. Failure to learn and fact-check in this simplified scenario has implications for pedagogies for fact-checking LLMs that require a more complex integration of skills.

We predicted that humans would learn more from a virtual student than from reading the text alone and that humans who corrected the erroneous virtual student mistakes would learn the most. These predictions follow from the literature on reading comprehension, mental models, and depth of processing [7, 11, 17]. Because both erroneous and correct virtual students enacted reading comprehension strategies, both virtual students guide the human participant in constructing a mental model of the text. The erroneous student, by giving the human opportunities to check and correct facts against the source text, should engage the deepest processing of the text. This is the same mechanism for learning we expect in fact-checking an LLM: students who critically evaluate claims and facts generated by the LLM, and correct them, should learn more from that experience because they are engaging in deeper processing.

The results of our two experiments do not fully support the fact-checking hypothesis. In Experiment 1 with undergraduate participants, there was zero learning except in the correct virtual student condition (Cor). Undergraduate participants were only 1.29 times as likely to change the majority of concept maps in the erroneous virtual student condition (Err) compared to the Cor condition, with 50 out of 92 participants failing to change the majority of maps in the Err condition. In Experiment 2 with crowd worker participants, there was learning in all conditions, but only the Err condition had significantly more learning than Read. Crowdworker participants were 49.20 times as likely to change

most of the concept maps in the Err as the Cor condition, with 83 out of 85 participants changing the majority of maps in the Err condition.

We believe that differences in correction behavior and learning across these two studies arise from differences in critical thinking engagement. The undergraduate participants, reading a text at approximately the 10th-grade reading level on the same subject as a course they were voluntarily taking, did not engage in critical thinking or its reading comprehension prerequisites. In contrast, the crowdworker participants, who were paid to participate, engaged in critical thinking and comprehension. Unfortunately, we cannot know for certain whether the difference in correction behavior and learning was solely one of motivation (which we suspect) or whether there could be some difference in ability between the two groups driving the difference in correction behavior and learning — this is perhaps the greatest limitation of our study. Paid compensation (or fear of not being paid) might be the reason, but these groups also differ in their educational attainment (crowdworkers were older and 40% had earned an undergraduate degree), and potentially differ in their familiarity with computers (crowdworkers may be more proficient). However, if the only participants who can learn from fact-checking are those with substantially higher ability or those who are substantially more motivated, then fact-checking is a niche pedagogy for high-ability students, rather than a general pedagogy to help students learn with LLMs.

Our experiments suggest that working with a virtual student that enacts reading comprehension strategies without making any mistakes, i.e. the Cor condition, is the most effective general pedagogy. The Cor condition was the only condition to have nonzero learning in both studies. For undergraduates, it was the only condition with learning, and for crowd workers, it was not significantly different from the Err condition. Thus the Cor condition worked both when participants put in little effort and when participants put in more effort. Altogether, these results suggest that fact-checking LLMs is not a reliable approach to learning content in a particular domain.

We did not evaluate the effectiveness of fact-checking on enhancing critical thinking skills — it may be that fact-checking can contribute to the development of these skills, as has been suggested by a study on debate [23]. However, in order for fact-checking practice to improve critical thinking skills, students must first actually engage in fact-checking, which we did not observe with undergraduate participants in Experiment 1. We suggest future research include critical thinking pre- and post-tests to measure this outcome. In summary, our results indicate that learning by correcting errors can be effective, but only if students put in the effort.

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